Problem-Solution Fit

Date	29/09/26
Team ID	SWTID-2026-5240
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Source-based conversion of the EduGenie documentation into the supplied canvas.

1. CUSTOMER SEGMENT(S) [CS]

- School / college students
- Beginners & self-learners
- Educational institutions
- Content platforms

6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL]

- Minimal hardware / low-footprint goal
- Limited prior knowledge or technical experience
- Need simple, accessible learning support
- Works well on devices such as Mac M1

5. AVAILABLE SOLUTIONS PROS & CONS [AS]

Existing educational resources / static content.

Pros: information is available.

Cons: content may feel overwhelming and may not combine instant Q&A;, quizzes, summaries and personalized paths.

2. PROBLEMS / PAINS + ITS FREQUENCY [PR]

- Complex concepts can be difficult to understand.
- Need quick academic answers.
- Long passages need concise revision.
- Need self-assessment and structured learning.

Frequency: recurring study needs; not quantified in source.

9. PROBLEM ROOT / CAUSE [RC]

- Traditional content may be overwhelming or inaccessible.
- Learners have different levels of prior knowledge.
- Support is needed across Q&A;, explanation, revision, assessment and planning.
- One-size-fits-all learning can limit personalization.

7. BEHAVIOR + ITS INTENSITY [BE]

- Ask questions when doubts arise.
- Request simpler explanations.
- Generate quizzes to check understanding.
- Summarize content for revision.
- Request beginner-to-advanced learning paths.

Intensity: recurring/task-driven; not quantified.

3. TRIGGERS TO ACT [TR]

- Difficult or unfamiliar topic
- Need for a quick answer
- Revision / self-assessment
- Long passage to summarize
- Starting a new topic / learning path

10. YOUR SOLUTION [SL]

EduGenie — lightweight AI learning assistant.

- Gemini 1.5 Pro: Q&A;, summaries, quizzes, learning paths.
- LaMini-Flan-T5-783M: simple explanations.
- FastAPI backend + HTML/CSS frontend.

8. CHANNELS OF BEHAVIOR [CH]

ONLINE

- EduGenie web interface
- FastAPI application
- Google Gemini API / cloud AI

OFFLINE

- Local Python/FastAPI environment
- Local lightweight explanation model

4. EMOTIONS BEFORE / AFTER [EM]

BEFORE

- Content feels complex / overwhelming
- Difficulty with unfamiliar concepts

AFTER

- Clearer understanding
- Concise answers / summaries
- Self-assessment and structured guidance

Note: Frequency/intensity are marked as not specified where the source does not quantify them.